

Office Linux Database Access Runbook (§25)

Target Audience: Office staff, founders, data analysts, and developers working on Linux workstations (Ubuntu, Debian, Fedora, Arch Linux, Linux Mint).

Database: PostgreSQL (AWS RDS / VPC Staging & Production Cluster)

Access Tier: Least-privilege Read-Only Analytics & Administrative Tunnels

1. Architecture Overview



Because the PostgreSQL database resides in a private VPC subnet without public IP exposure, all connections from outside the VPC must traverse an encrypted **SSH local port forwarding tunnel**.

2. Recommended Database Clients for Linux

Option A: DBeaver Community (Recommended for Analysts & Founders)

DBeaver is a powerful, free, universal database GUI that handles SSH tunnels natively.

- Installation (Ubuntu / Debian / Mint):

```
sudo snap install dbeaver-ce
# OR via APT:
sudo add-apt-repository ppa:serge-rider/dbeaver-ce -y
sudo apt update && sudo apt install dbeaver-ce -y
```

- Installation (Fedora / RHEL):

```
sudo flatpak install flathub io.dbeaver.DBeaverCommunity
```

Option B: TablePlus (Modern, Lightweight UI)

- Installation (Debian / Ubuntu):

```
# Add TablePlus GPG key and repo
wget -qO - https://deb.tableplus.com/apt.key | gpg --dearmor | sudo tee /etc/apt/trusted.gpg.d/tableplus.gpg
sudo add-apt-repository "deb [arch=amd64] https://deb.tableplus.com/debian/tableplus tableplus"
sudo apt update && sudo apt install tableplus -y
```

Option C: `psql` CLI (Fast Terminal Access)

- Installation:

```
sudo apt install postgresql-client -y # Debian/Ubuntu
sudo dnf install postgresql -y        # Fedora
```

3. Establishing the SSH Tunnel

Step 3.1: SSH Key Setup

Ensure your private key has strict POSIX permissions:

```
chmod 600 ~/.ssh/networkpeer_bastion_key.pem
```

Step 3.2: Configure SSH Host Shortcut (`~/.ssh/config`)

Add the following configuration to `~/.ssh/config` :

```
Host networkpeer-bastion
  HostName ec2-13-53-128-44.eu-north-1.compute.amazonaws.com
  User ubuntu
  IdentityFile ~/.ssh/networkpeer_bastion_key.pem
  ServerAliveInterval 60
  ServerAliveCountMax 3
  LocalForward 5433 networkpeer-staging-db.c7x8y9z0.eu-north-1.rds.amazonaws.com:5432
```

Step 3.3: Launching the Tunnel (One-Liner)

Run:

```
ssh -N networkpeer-bastion
```

(The `-N` flag tells SSH not to execute a remote shell, keeping the tunnel open silently in the background).

4. Persistent Auto-Reconnecting Tunnel (Systemd Service)

To keep the database tunnel automatically connected on your office Linux workstation without needing a manual terminal:

- 1. Create `/etc/systemd/system/networkpeer-db-tunnel.service` :

```
[Unit]
Description=NetworkPeer PostgreSQL SSH Tunnel
After=network.target

[Service]
User=aditya
ExecStart=/usr/bin/ssh -NT -o ServerAliveInterval=60 -o ExitOnForwardFailure=yes -L 5433:net
Restart=always
RestartSec=10

[Install]
WantedBy=multi-user.target
```

- 2. Enable and start the service:

```
sudo systemctl daemon-reload
sudo systemctl enable networkpeer-db-tunnel
sudo systemctl start networkpeer-db-tunnel
sudo systemctl status networkpeer-db-tunnel
```

5. Least-Privilege Read-Only User Setup

To prevent accidental data alterations or destructive drops, data analysts and office workstations must connect using a dedicated read-only role:

```
-- Run as superuser/admin on staging PostgreSQL:
CREATE ROLE networkpeer_readonly WITH LOGIN PASSWORD 'SecuredReadOnlyPass2026!';
GRANT CONNECT ON DATABASE networkpeer_staging TO networkpeer_readonly;
GRANT USAGE ON SCHEMA public TO networkpeer_readonly;
GRANT SELECT ON ALL TABLES IN SCHEMA public TO networkpeer_readonly;
ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT SELECT ON TABLES TO networkpeer_readonly;
```

6. Connecting via DBeaver / TablePlus

Configure your connection settings in your GUI tool:

Parameter	Value
Host	127.0.0.1 (or localhost)
Port	5433 (maps through tunnel to RDS 5432)

Parameter	Value
Database	networkpeer_staging
Username	networkpeer_readonly (or admin account)
Password	(As provisioned above)
SSL Mode	prefer or require

Test connection: Click **Test Connection** in DBeaver/TablePlus -> Status should display **Connected** (PostgreSQL 16.x) .

7. Useful Diagnostic Queries for Founders & Analysts

A. Monitor Unfunded vs. Funded Jobs

```
SELECT
  id,
  title,
  status,
  funding_status,
  (budget_cents / 100.0) AS budget_inr,
  created_at
FROM jobs
ORDER BY created_at DESC
LIMIT 25;
```

B. Worker Review Queue & Correctionist Eligibility

```
SELECT
  u.id AS user_id,
  u.full_name,
  u.email,
  u.mobile_number,
  wp.is_available,
  wp.eligible_roles
FROM users u
LEFT JOIN worker_profiles wp ON u.id = wp.user_id
WHERE u.role = 'WORKER';
```

C. Evidence Quality & OCR Verification Audit

```
SELECT
  sub.id,
  sub.job_id,
  sub.status,
```

```
sub.ocr_confidence,  
sub.detected_script,  
sub.created_at  
FROM submissions sub  
ORDER BY sub.created_at DESC  
LIMIT 50;
```

8. Troubleshooting Common Linux Workstation Issues

1. Error: `bind: Address already in use` :

- Cause: Another process or previous tunnel is bound to port `5433` .
- Fix:

```
sudo lsof -i :5433  
kill -9 <PID>
```

2. Error: `Permissions 0644 for id_rsa are too open` :

- Fix: `chmod 600 ~/.ssh/networkpeer_bastion_key.pem`

3. Local PostgreSQL collision on 5432:

- Always map the local tunnel port to `5433` instead of `5432` to avoid conflicts if a local PostgreSQL daemon is installed on your Linux machine.